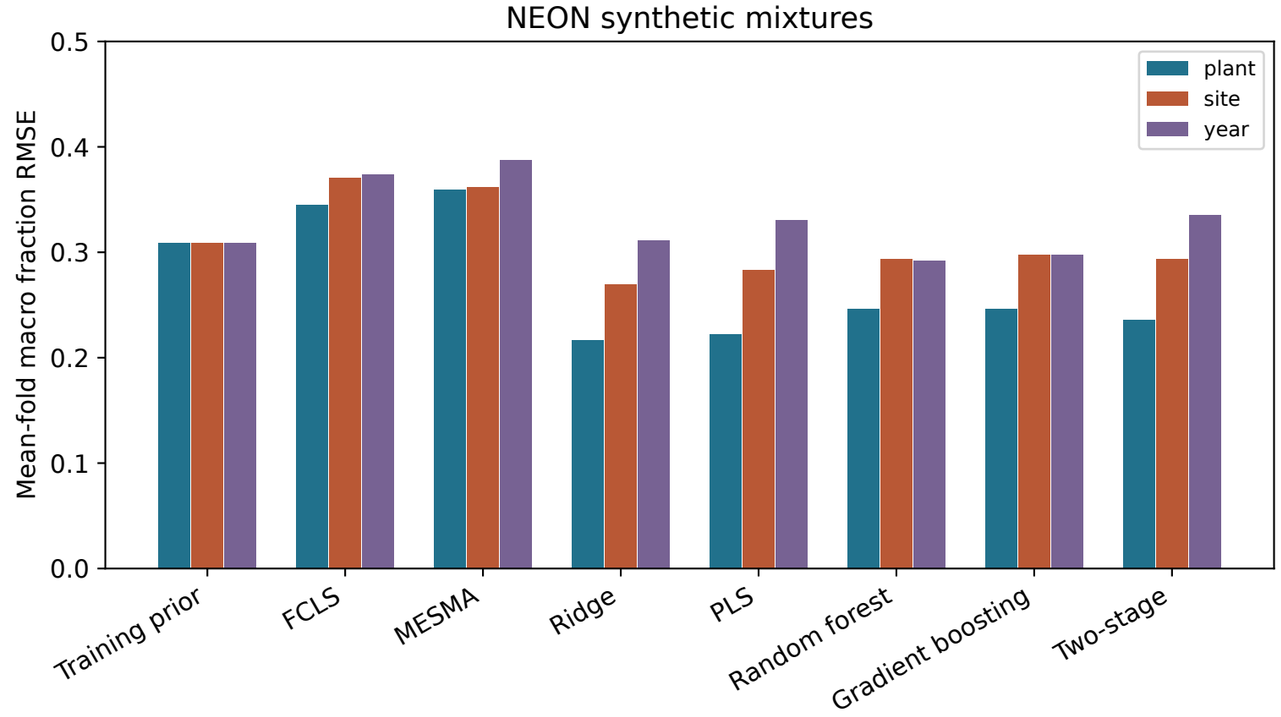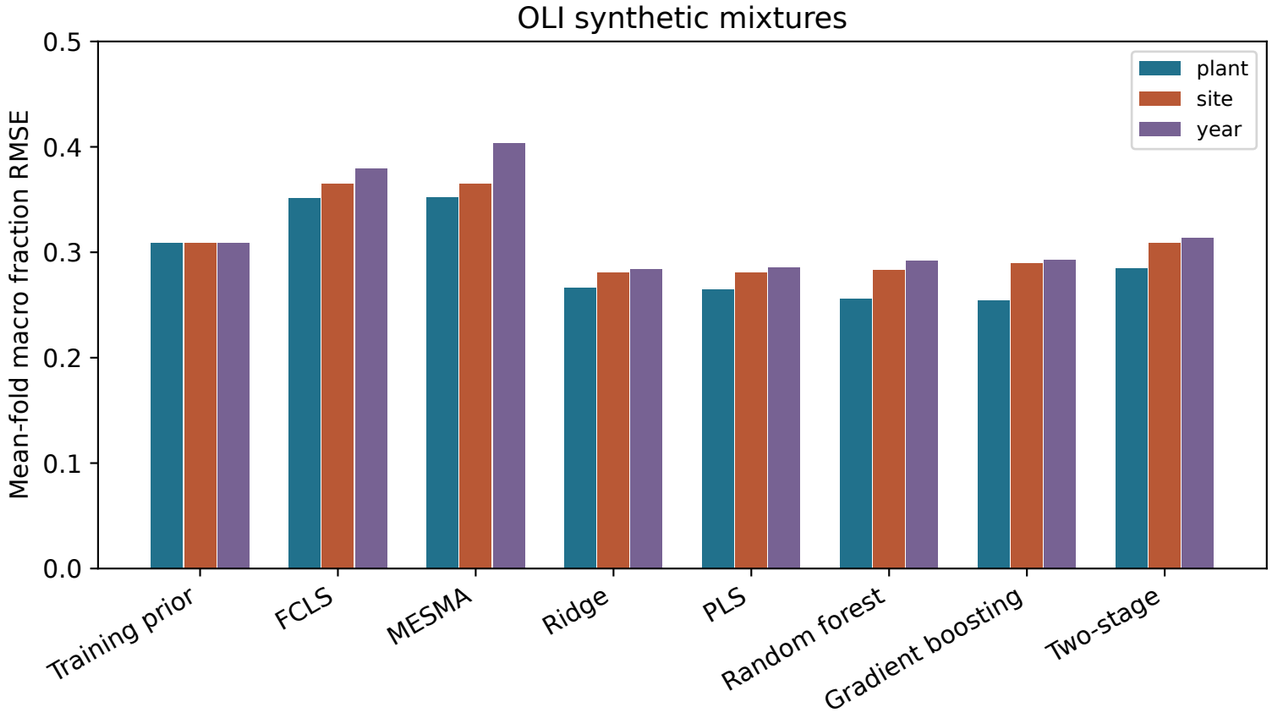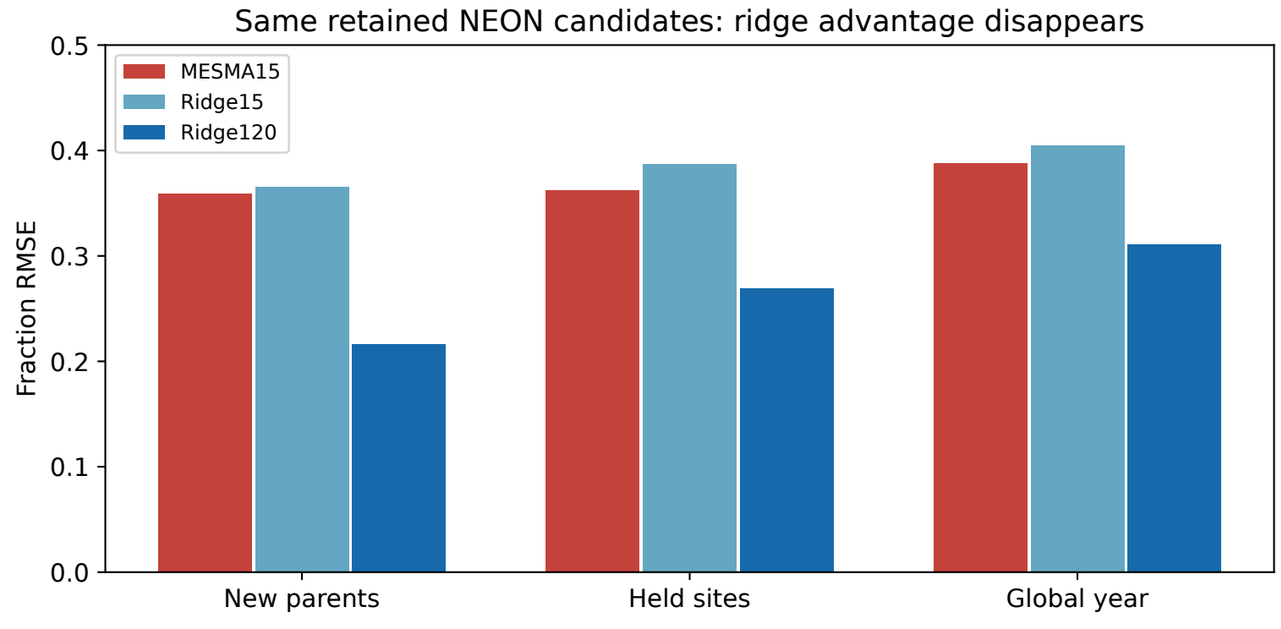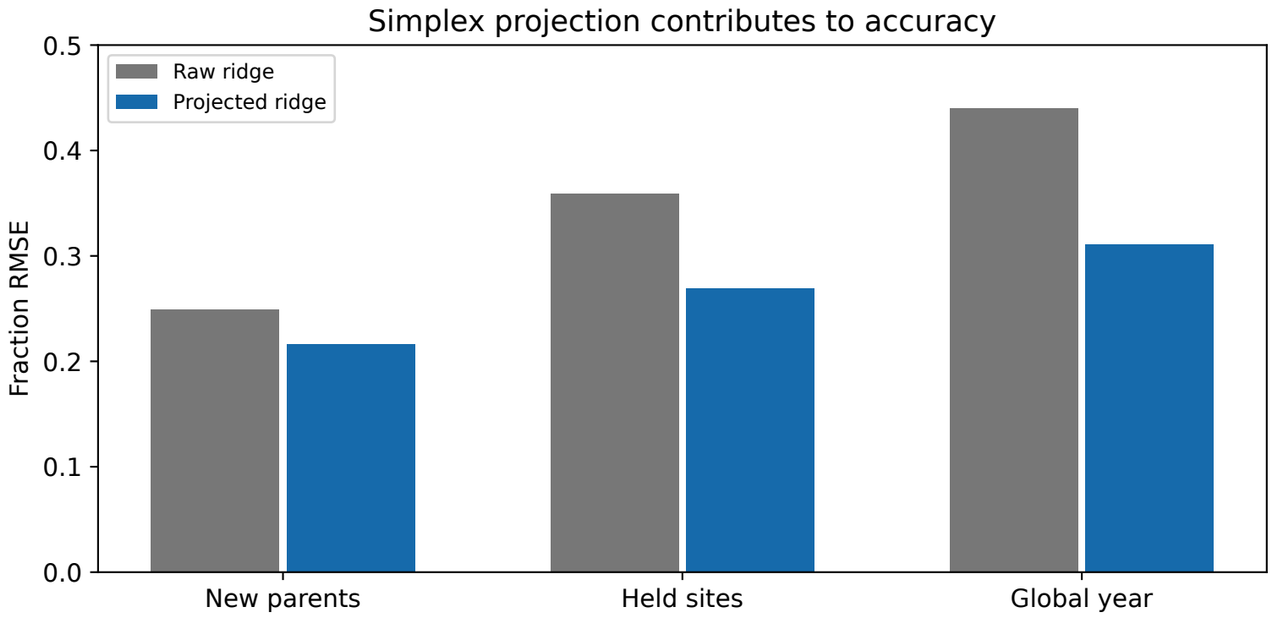Synthetic abundance gains depend on domain, priors and retained library